

Anindya Mondal

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Education

Surrey Institute for People-centred AI, CVSSP, University of Surrey <i>PhD Candidate in Artificial Intelligence</i> Research Focus: Integrating Auxiliary Information for Representation Learning in Natural World	Guildford, United Kingdom Oct 2022 – Present
Jadavpur University <i>B.E. in Electronics and Telecommunication Engineering (Hons.); GPA: 8.79/10</i>	Kolkata, India Aug 2018 – Jun 2022

Research Focus

2023 – 2024: Vision-Language Integration, Multimodal Learning, Action Recognition, Text-to-Image Synthesis, Object Counting, and 3D/4D Generation.

2020 – 2022: Graph Neural Networks, Time-series Analysis, Graph Signal Processing, Neuromorphic Vision Systems, and Subspace Learning Techniques.

Publications

AAAI '25: Anindya Mondal, Sauradip Nag, Xiatian Zhu, Anjan Dutta, “OmniCount: Multi-label Object Counting with Semantic-Geometric Priors,” DOI: 10.48550/arXiv.2403.05435.

ICCVW '23: Anindya Mondal, Sauradip Nag, Joaquin M Prada, Xiatian Zhu, Anjan Dutta, “Actor-agnostic Multi-label Action Recognition with Multi-modal Query,” DOI: 10.1109/ICCVW60793.2023.00086.

ICASSP '23: JAC Correa, JH Giraldo, Anindya Mondal, et al., “Time-varying Signals Recovery via Graph Neural Networks,” DOI: 10.1109/ICASSP49357.2023.10096168.

EUSIPCO'22: Anindya Mondal, et al., “Recovery of Missing Sensor Data by Reconstructing Time-varying Graph Signals,” DOI: 10.23919/EUSIPCO55093.2022.9909940.

ICCVW '21: Anindya Mondal, R Shashant, et al., “Moving Object Detection for Event-based Vision using Graph Spectral Clustering,” DOI: 10.1109/ICCVW54120.2021.00103.

Research Experience

Surrey Institute for People-centred AI, CVSSP, University of Surrey <i>Doctoral Researcher</i>	Guildford, UK Oct 2022 – Present
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- Developed a diffusion-based text-to-image generation model for high-quality exemplar generation aimed at object counting.
- Created a class-agnostic object counting model utilizing semantic and geometric priors, enhancing system adaptability.
- Designed and implemented a transformer-based multimodal action recognition model, boosting recognition accuracy by 50%.
- Led a project to establish a benchmark for animal action recognition, including detection and segmentation tasks.

Indian Institute of Science

Research Intern

Bengaluru, India

May 2022 – Aug 2022

- Innovated a source-free domain adaptation method for image classification, improving robustness across different target domains.

Jadavpur University

Undergraduate Research Assistant

Kolkata, India

Oct 2020 – May 2022

- Implemented a Sobolev norm minimization technique for reconstructing time-varying graph signals, reducing errors significantly.
- Developed a semi-supervised learning framework for semantic segmentation using graph theory, demonstrating improved efficiency.
- Enhanced detection of moving objects from event data using an adapted graph spectral clustering approach.

Technical Skills

Programming: Proficient in Python, MATLAB, C.

Libraries: Experienced with PyTorch, TensorFlow, Scikit-Learn, NumPy, SciPy, Pandas.

Tools: Proficient in Git, LaTeX, Jupyter Notebook, and Docker.

Awards and Honors

2024: Awarded the AAAI 2025 Conference cum Travel Grant (worth \$1200), Philadelphia, USA.

2023: Awarded the ICCV 2023 Conference Grant, Paris, France.

2022: Awarded the Postgraduate Studentship at the University of Surrey, UK.

2022: Recipient of the Uplink Research Internship Award from ACM SIGKDD India Chapter.

Professional Experience

Teaching.....

2023 – 2025: Teaching Assistant for Applied Machine Learning (EEEM068), Advanced Topics in Computer Vision and Deep Learning (EEEM071) at the University of Surrey and UKRI Central for Doctoral Training (CDT).

Peer Review.....

2022 – 2025: Peer reviewer for ICASSP, ICCV, CVPR, ECCV, NeurIPS, ICPR, ICASSP, Transactions on Signal Processing (TSP), Transactions on Signal and Information Processing over Networks (TSIPN).